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**Report of Mini-Projet :**  
**Analysis of the Karate Club Network**

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# Contents

|                                                   |          |
|---------------------------------------------------|----------|
| <b>1 Graph Modeling</b>                           | <b>2</b> |
| 1.1 Overview of the Karate Club Network . . . . . | 2        |
| 1.2 Type of Graph . . . . .                       | 2        |
| 1.3 Graph Construction . . . . .                  | 2        |
| 1.4 Adjacency Matrix . . . . .                    | 3        |
| 1.5 Network Visualization . . . . .               | 3        |
| <b>2 Structural Analysis</b>                      | <b>4</b> |
| 2.1 Graph Order and Size . . . . .                | 4        |
| 2.2 Degree Distribution . . . . .                 | 4        |
| 2.3 Clustering Coefficient . . . . .              | 4        |
| 2.4 Frequent Motifs and Cliques . . . . .         | 5        |
| 2.5 k-Cores . . . . .                             | 5        |
| <b>3 Centrality Analysis</b>                      | <b>5</b> |
| 3.1 Degree Centrality . . . . .                   | 5        |
| 3.2 Betweenness Centrality . . . . .              | 5        |
| 3.3 Closeness Centrality . . . . .                | 6        |
| 3.4 Eigenvector Centrality . . . . .              | 6        |
| <b>4 Centrality and Bridge Node Analysis</b>      | <b>7</b> |
| 4.1 Bridge Nodes . . . . .                        | 7        |
| 4.2 Visualization . . . . .                       | 7        |
| 4.3 Interpretation . . . . .                      | 7        |

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# 1 Graph Modeling

## 1.1 Overview of the Karate Club Network

The *Karate Club* network represents friendship relationships among the members of a martial arts club studied by Zachary. Each node corresponds to a club member, and each edge represents a friendship connection between two members.

## 1.2 Type of Graph

The graph is characterized as follows:

- **Undirected:** A friendship is mutual — if member  $A$  is a friend of member  $B$ , then  $B$  is also a friend of  $A$ .
- **Unweighted:** All relationships have the same importance. No numerical value (or “weight”) is assigned to the edges.

## 1.3 Graph Construction

There are two main approaches to build the Karate Club network graph:

1. **Manual creation of the adjacency matrix:** One could directly define the  $34 \times 34$  adjacency matrix  $A$  by hand, where each entry  $A_{ij}$  is 1 if there is a friendship between members  $i$  and  $j$ , and 0 otherwise. However, this approach is tedious and error-prone, especially for larger graphs.
2. **Building from an edge list:** A simpler and less error-prone approach is to define the list of edges (pairs of connected nodes) and then generate the adjacency matrix programmatically using Python. Since the Karate Club network contains 34 nodes, the resulting adjacency matrix will be of size  $34 \times 34$ .

We chose the second approach for its efficiency and accuracy. Each node represents a club member, and each edge represents a friendship relation between two members.

Below is the Python code used to construct the adjacency matrix and the undirected graph:

```
import numpy as np
import networkx as nx

# Number of nodes
n = 34

# Initialize an empty adjacency matrix
A = np.zeros((n, n), dtype=int)

# Define edges (friendship relations)
edges = [
    (0,1),(0,2),(0,3),(0,4), ... , (31,32),(31,33),(32,33)
]

# Fill the adjacency matrix symmetrically
for i, j in edges:
    A[i, j] = 1
    A[j, i] = 1 # make it symmetric

# Create the graph using NetworkX
G = nx.Graph()
G.add_nodes_from(range(n))
G.add_edges_from(edges)
```

## 1.4 Adjacency Matrix

The adjacency matrix  $A$  for the Zachary Karate Club network represents the connections (friendships) between the 34 members. Each element  $A_{ij}$  equals 1 if there is a friendship between member  $i$  and member  $j$ , and 0 otherwise. Because the graph is undirected, the matrix is symmetric:  $A_{ij} = A_{ji}$ .

Below is a representative  $10 \times 10$  section of the full  $34 \times 34$  adjacency matrix:

$$A_{10 \times 10} = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

This matrix highlights the dense interconnections among the first few members of the club, especially within the initial subgroup centered around the instructor and the president.

## 1.5 Network Visualization

To better illustrate the structure of the Karate Club network, the figure below shows the undirected, unweighted graph generated from the adjacency matrix. Nodes represent members, and edges represent mutual friendships.

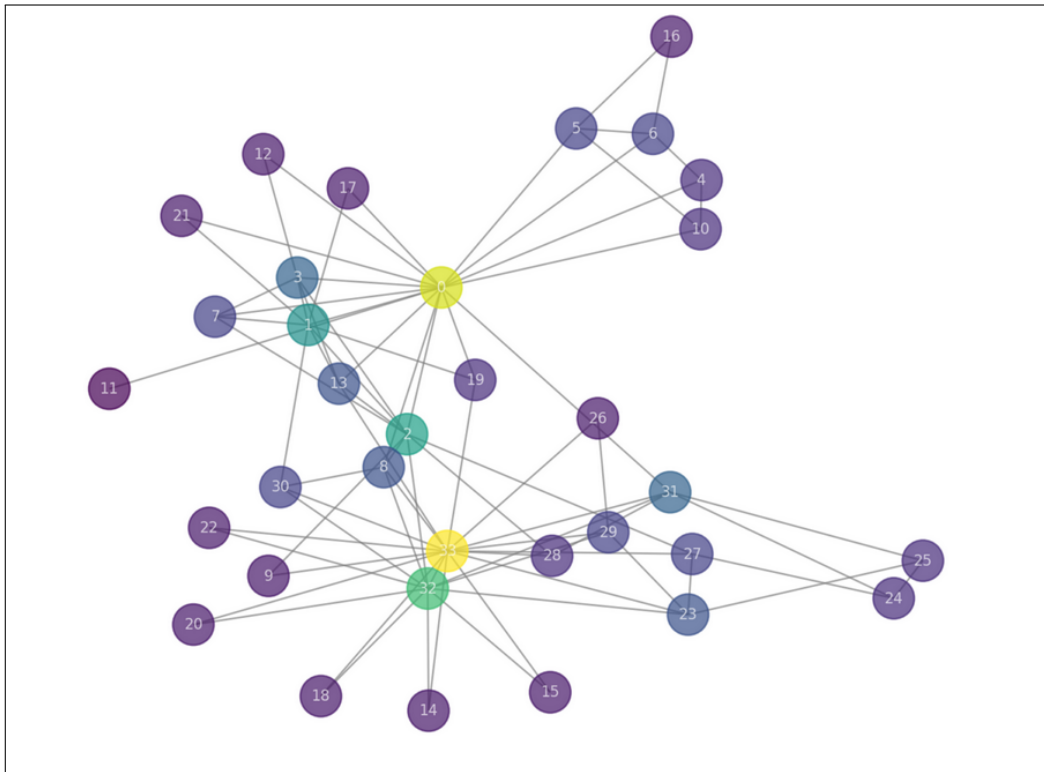


Figure 1: Graph

## 2 Structural Analysis

### 2.1 Graph Order and Size

The Karate Club network has 34 nodes (members) and 75 edges (friendship links). The graph is moderately connected — not everyone knows everyone, but most members are part of the same friendship structure. This shows a medium-density social network.

### 2.2 Degree Distribution

The degree of a node is the number of direct friends a member has. Most members have only a few friends, while a small number are very well connected. The distribution is as follows:

| Degree | Number of Nodes |
|--------|-----------------|
| 1      | 1               |
| 2      | 12              |
| 3      | 6               |
| 4      | 6               |
| 5      | 3               |
| 6      | 1               |
| 8      | 1               |
| 10     | 1               |
| 12     | 1               |
| 15     | 1               |
| 17     | 1               |

The most connected members (degrees 15 and 17) are likely the leaders or central figures of the club.

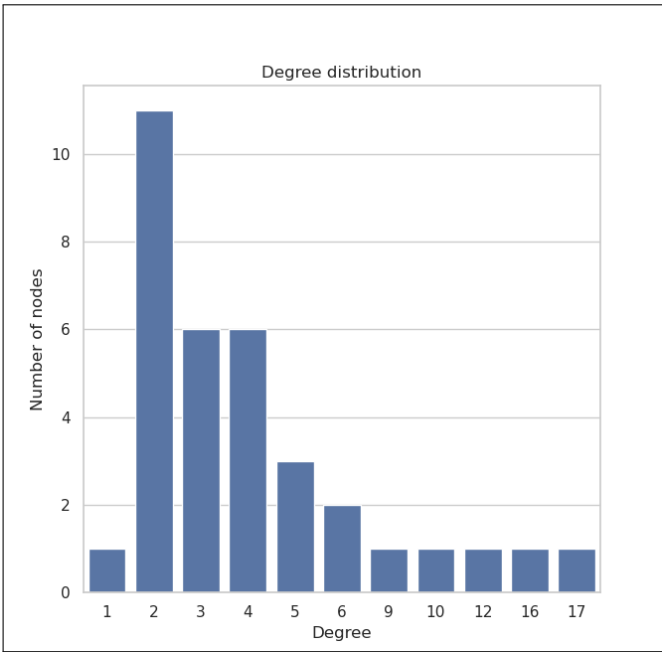


Figure 2: Enter Caption

### 2.3 Clustering Coefficient

The average clustering coefficient is 0.573. This means members often form small, tightly connected groups of friends — showing strong local cohesion within the club.

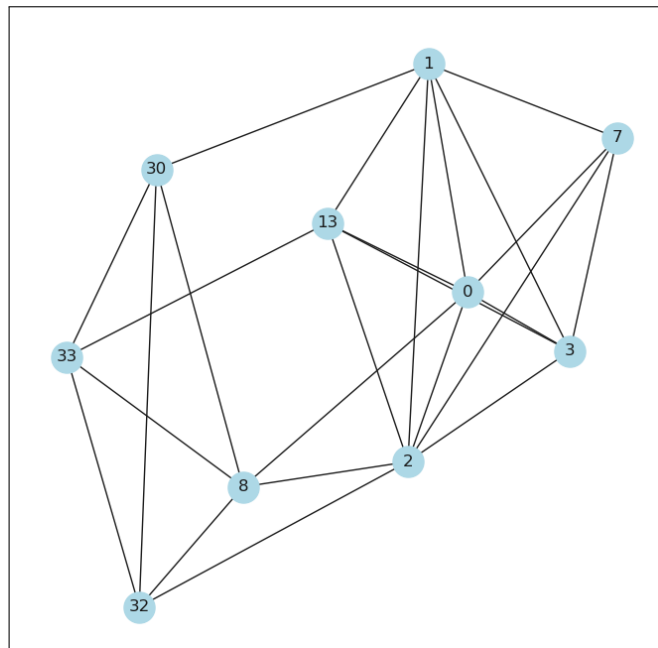


Figure 3: 4-cores Graph

## 2.4 Frequent Motifs and Cliques

The network has 44 triangles and 35 cliques, with the largest clique containing 5 members. These small groups represent circles of friends who all know each other, showing how social clusters form in the club.

## 2.5 k-Cores

The maximum k-core of the network is 4. This means there is a dense inner group where each member has at least 4 connections. These core members form the social center of the club, while others are more peripheral.

# 3 Centrality Analysis

## 3.1 Degree Centrality

Degree centrality measures how many direct connections each member has. The top 5 most connected members are:

| Node | Degree Centrality |
|------|-------------------|
| 33   | 0.515             |
| 0    | 0.455             |
| 32   | 0.364             |
| 2    | 0.303             |
| 1    | 0.242             |

Nodes 33 and 0 have the highest degree centrality, meaning they are the most connected members in the network — likely the main leaders of the club.

## 3.2 Betweenness Centrality

Betweenness centrality identifies members who act as bridges between different groups. The top nodes are:

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| Node | Betweenness Centrality |
|------|------------------------|
| 0    | 0.415                  |
| 33   | 0.343                  |
| 2    | 0.218                  |
| 32   | 0.155                  |
| 8    | 0.101                  |

Nodes 0 and 33 serve as key connectors between subgroups, showing they play important roles in maintaining communication within the club.

### 3.3 Closeness Centrality

Closeness centrality measures how quickly a member can reach others in the network. The top nodes are:

| Node | Closeness Centrality |
|------|----------------------|
| 2    | 0.559                |
| 33   | 0.550                |
| 0    | 0.524                |
| 8    | 0.516                |
| 13   | 0.516                |

These members can easily reach others, meaning they are centrally positioned and can spread information quickly.

### 3.4 Eigenvector Centrality

Eigenvector centrality measures overall influence — a member connected to other important members will have a high score. The top nodes are:

| Node | Eigenvector Centrality |
|------|------------------------|
| 33   | 0.387                  |
| 0    | 0.340                  |
| 2    | 0.329                  |
| 32   | 0.318                  |
| 1    | 0.249                  |

Nodes 33 and 0 again stand out as the most influential individuals in the network. This aligns with the real-world observation that the two main leaders of the club (the instructor and the president) had the strongest social influence.

## Visualization

The figure below shows a bar plot comparing the top 5 nodes across all four centrality measures. **We can see from the graph below that member 0 and 33 have the most centrality between all member which are more likely to be the leaders of the two groups**

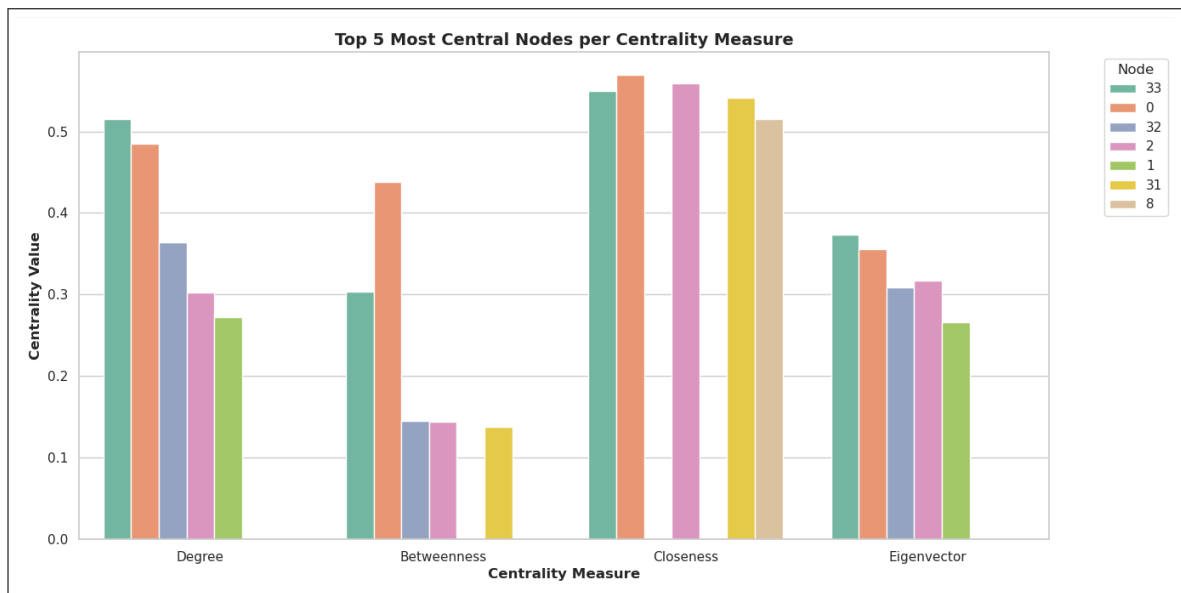


Figure 4: Centrality Graph

## 4 Centrality and Bridge Node Analysis

We analyzed the **Zachary's Karate Club** network to identify the most influential members and the nodes that connect both groups.

### 4.1 Bridge Nodes

Bridge nodes are those that connect the two factions. They ensure communication between both groups. In our network, bridge nodes include:

$$\{0, 1, 2, 5, 6, 8, 9, 13, 16, 19, 27, 28, 30, 31, 32, 33\}$$

### 4.2 Visualization

A simple bar chart was plotted using Seaborn to display the top 5 nodes by centrality values.

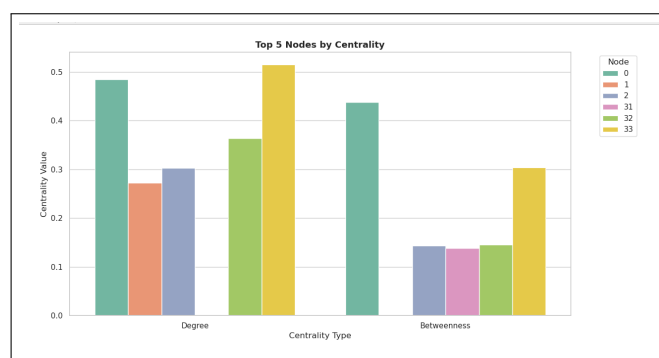


Figure 5: Centrality Withing Groups

### 4.3 Interpretation

Nodes 0 and 33 have the highest centrality values, meaning they are the most influential members in their respective groups. Bridge nodes play a key role in maintaining connections between the two factions, explaining how the network split during the conflict.